



OKinsurance

Shark Attack Risk Analysis

Insurance Product Proposal for Okinsurance

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Project Overview

Dataset: Global Shark Attack File (GSAF) — 7,103 records

Business Case: Research team at Okinsurance proposing a water-sport risk insurance product for US hotels & resorts.



Shark Interaction
Insurance



Standard Swim
Insurance

Hypotheses



H1 Risk is concentrated in a small number of countries



H2 Risk varies significantly by month / season



H3 Unprovoked attacks dominate — environmental, not behavioral, risk

Market Context — Where We Fit

Water-sport liability insurance already exists (e.g. open-water swim event coverage, water-sports business liability) — but it's *generic*, not tied to historical incident data.

Existing providers: Sadler Sports, XINSURANCE, American Visitor Insurance — general liability, not risk-segmented.



Our differentiator: no existing provider prices coverage using shark-specific historical incident data. That's Okinsurance's opportunity.



Data Cleaning — Dates

Challenge: the Date column was only partially auto-parsed by pandas on import — some rows kept full datetime values, others stayed as raw text (e.g. "23rd June").

Techniques applied:

- Extracted Month directly from correctly auto-parsed dates
- Regex recovery of month names from raw text on the remaining rows
- Selective null handling — dropna() applied only where needed

86%

of dates auto-parsed correctly on first read

12.4%

missing rate remaining after regex recovery



Data Cleaning — Country, Activity & Type



Country

Mixed casing, question marks, region/ocean names → standardized to title case, invalid entries grouped as "Unknown"



Activity

1,613 free-text variants (typos, combined descriptions) → mapped to 20 core categories

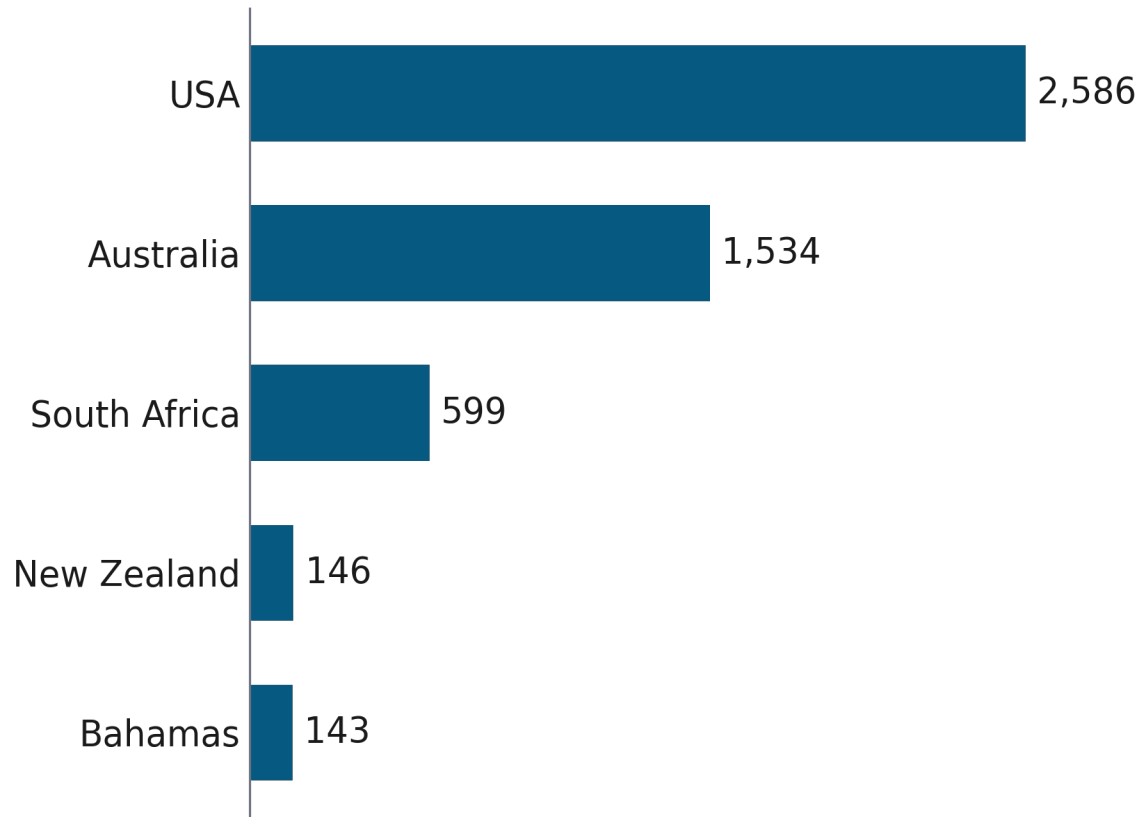


Type

Spelling/casing variants of Unprovoked / Provoked / Invalid → standardized labels, rare types grouped as "Other"

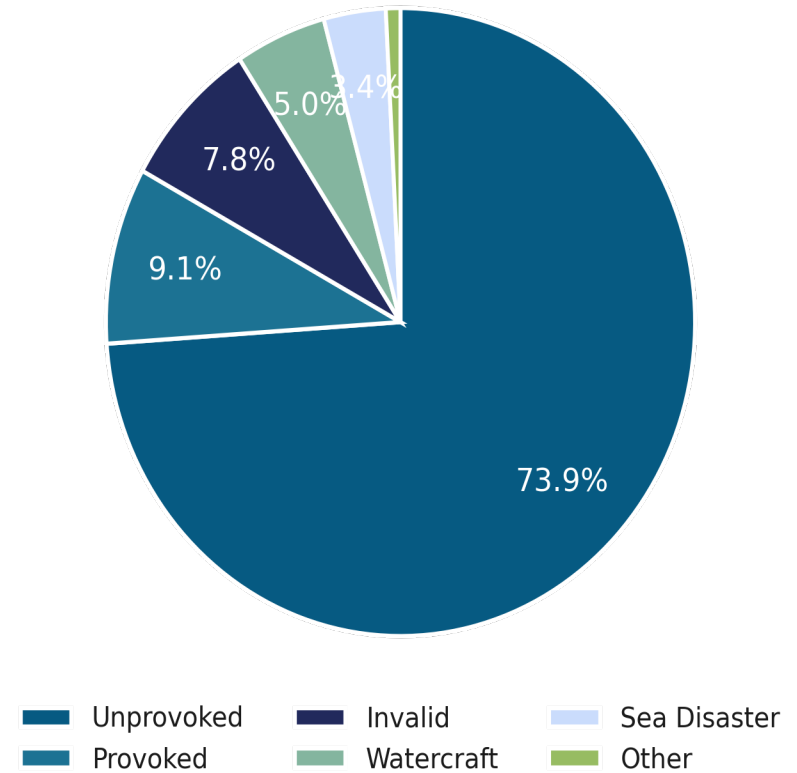
EDA — Geographic & Attack-Type Patterns

Top 5 Countries by Incident Count



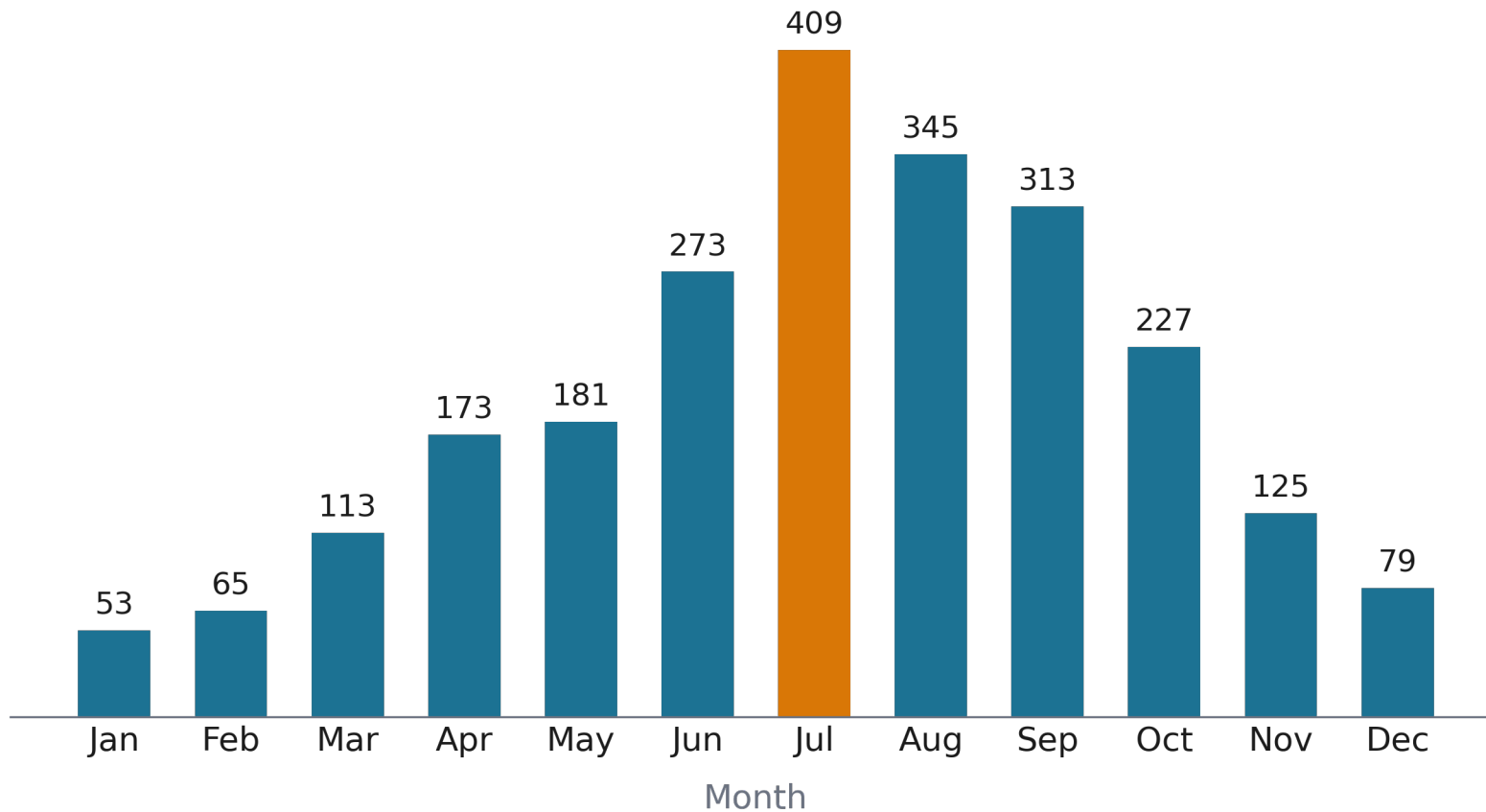
Bars show total number of recorded incidents per country (1900–present).

Attack Type Distribution (%)



Values are the % share of each attack type among all classified incidents.

EDA — Seasonality of Incidents in the USA

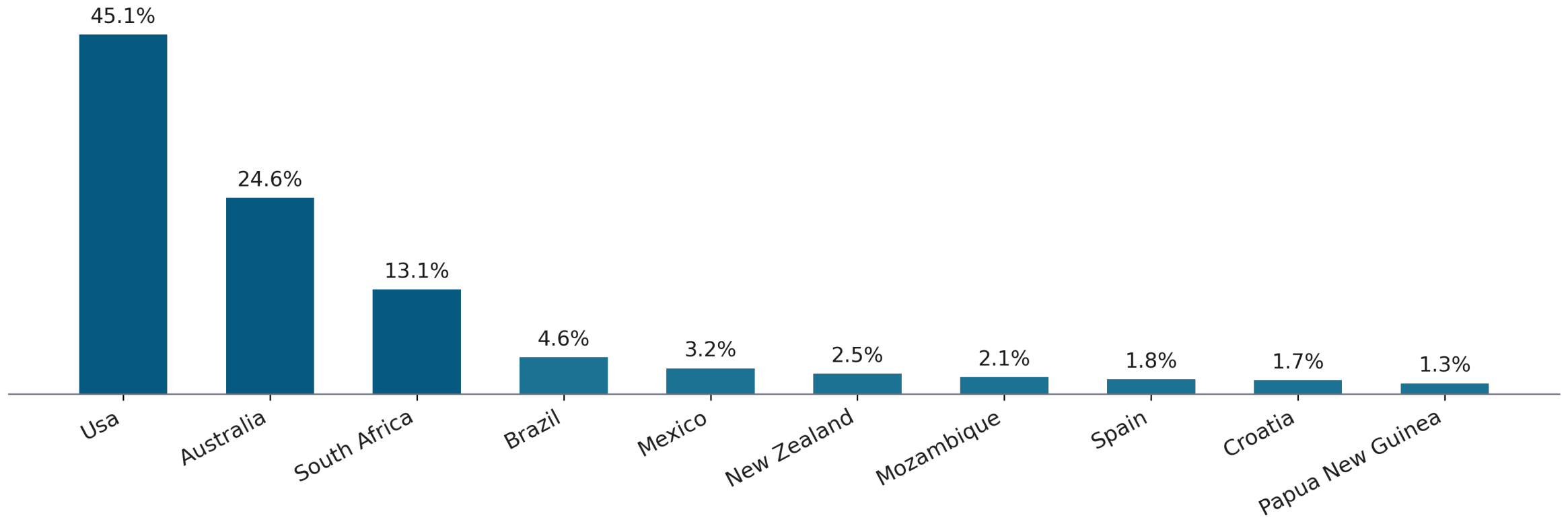


Peak: July

409 incidents

Clear seasonal pattern supports a seasonal premium loading for peak summer months.

EDA — Swimming-Specific Risk by Country



% share of swimming-specific incidents by country — USA, Australia & South Africa account for ~83% combined, directly informing Standard Swim Insurance pricing priority.



Major Obstacle

January showed an implausible 812 incidents — nearly double every neighboring month.

Lesson: question suspicious results before trusting them.

41%

of January rows had the exact date "Jan 1st" — a signature of incomplete original dates (year only), defaulted by Excel/pandas.

~~812~~ → 478

Corrected January incident count



Conclusion & Insights



H1 Supported — USA, Australia & South Africa dominate incident volume



H2 Supported — clear seasonal peak in July-August



H3 Supported — 73.9% of attacks are unprovoked (environmental, not behavioral, risk)



Proposal: a two-tier insurance product for US hotels & resorts — Shark Interaction and Standard Swim — with a seasonal premium adjustment recommended to Sales & Finance for peak summer months.

Business Recommendation

Based on our findings, we recommend a two-tier product for US hotels & resorts:



Shark Interaction Insurance

- For guests seeking active shark-encounter / diving experiences
- Higher-risk activity liability coverage
- Premium pricing tier — reflects elevated exposure



Standard Swim Insurance

- For guests swimming at the hotel's beach — no interaction sought
- Baseline liability coverage
- Standard pricing tier — lower relative risk



Seasonal loading: increase premiums across both tiers during peak incident months (July–August), per H2.



Rollout priority: USA, Australia & South Africa — highest incident concentration and swimming-specific risk share.



Final pricing left to Sales & Finance — this analysis defines risk segmentation, not dollar figures.



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Thank You!